DMP

A9126 DAC

Test Report

DMP Electronics Inc.

http://www.dmp.com.tw

Revision History

REV.	Date	History
REV 01	2015/03/19	1. Initial Version.
REV 02	2015/08/28	1. Initial Version after PCB modification.
REV 03	2015/09/10	1. Re-test by Tektronix Scope.

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1. Test Environment

Pixel Clock : 148.500MHz
Scope: Tektronix MSO 70604C-6GHz
Probe#1&2: Tektronix Probe P7340A
Probe#3: Tektronix Probe P6248
PCB: VDX3-6790

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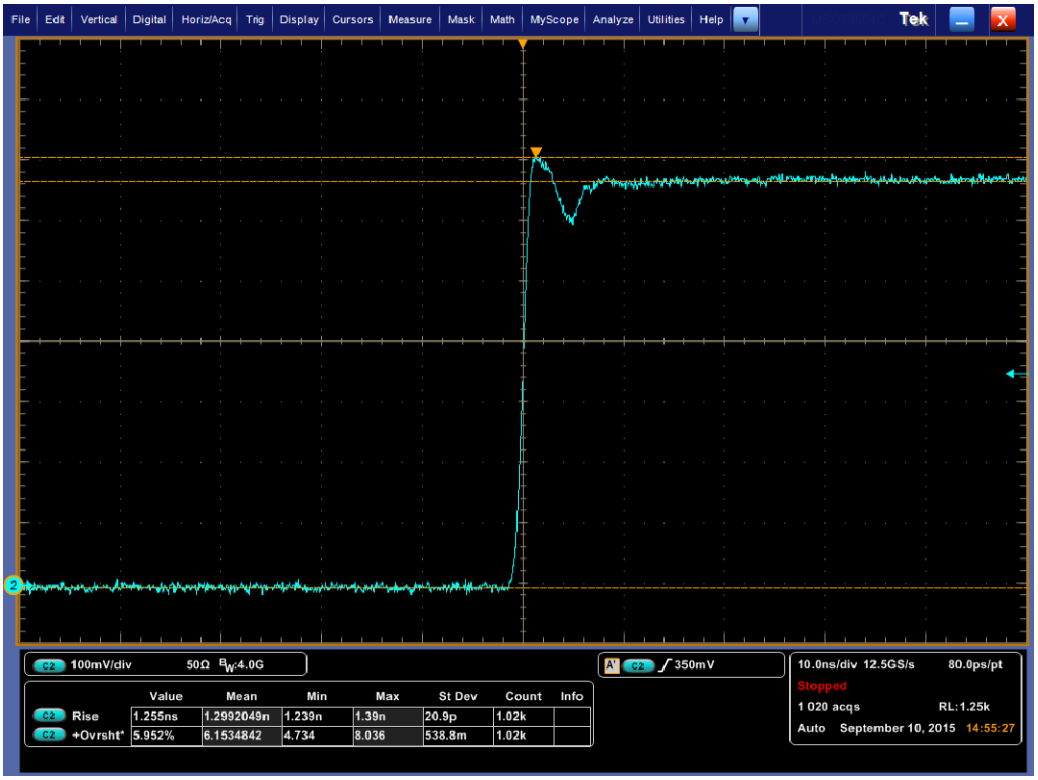
2. Analog Rising Time

Rising Time Table

10%~90%	PASS LIMITS	Mean	Minimum	Maximum	Result
Red	<=1.68ns	1.29ns	1.23ns	1.39ns	**Pass**
Green	<=1.68ns	1.24ns	1.17ns	1.33ns	**Pass**
Blue	<=1.68ns	1.28ns	1.21ns	1.39ns	**Pass**

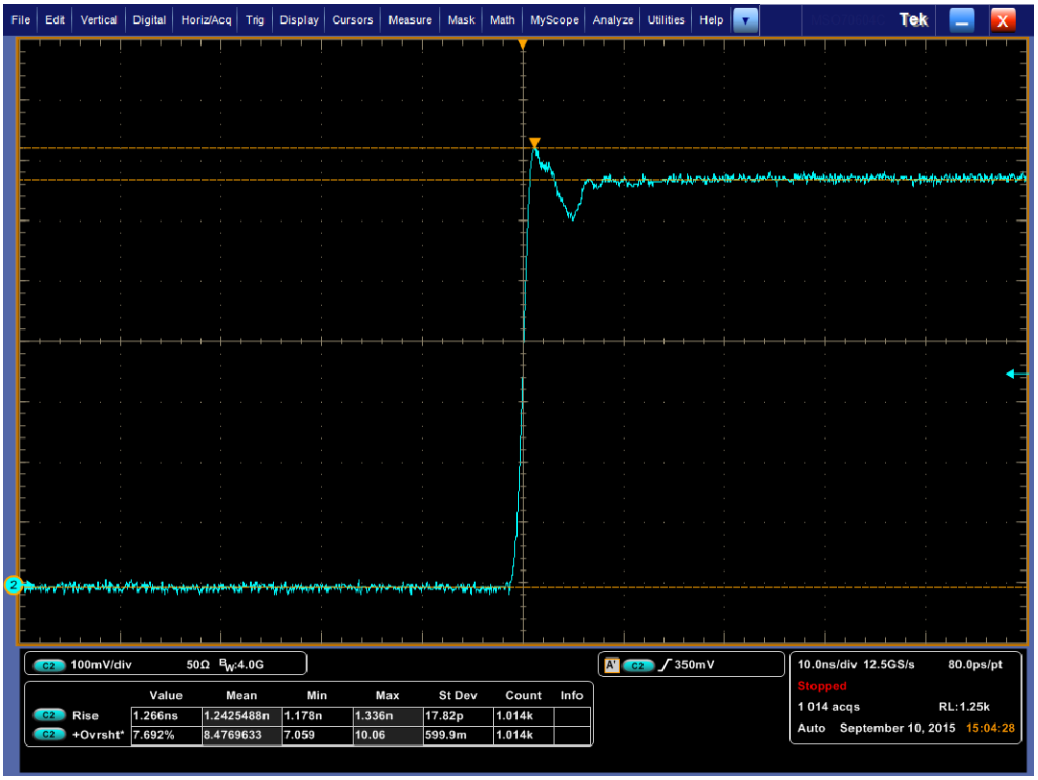
Pass Limits => 1.68ns = (1/148.500M)*25%

Analog Red Channel Rising Time

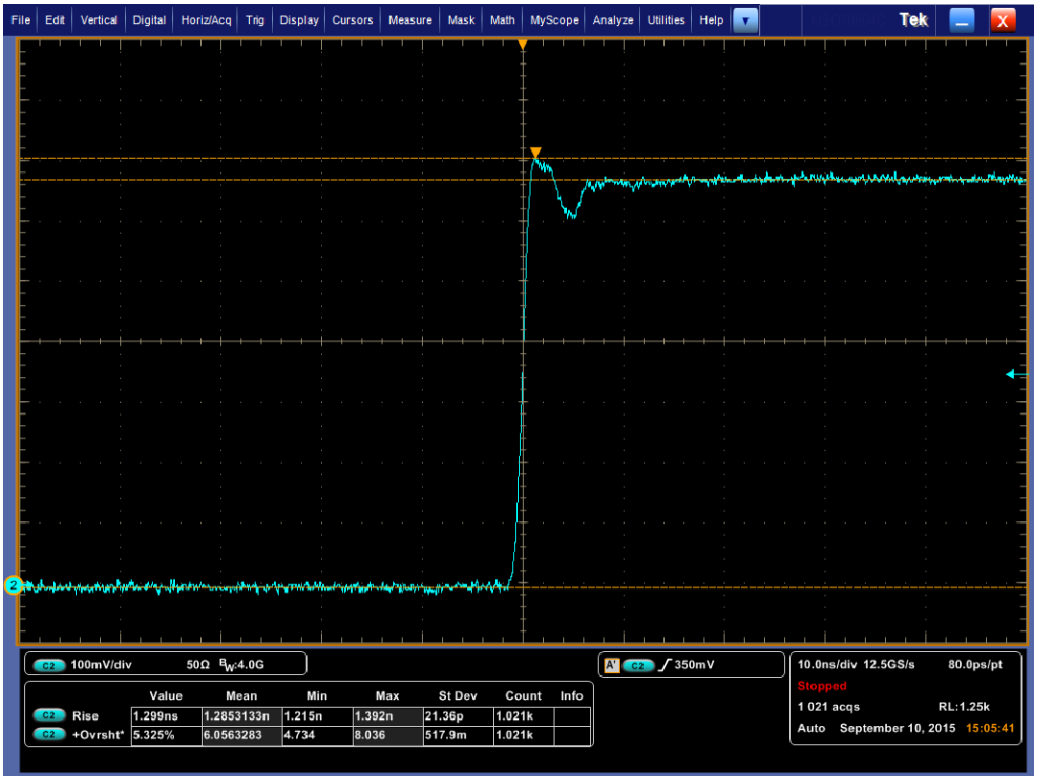


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Analog Green Channel Rising Time



Analog Blue Channel Rising Time



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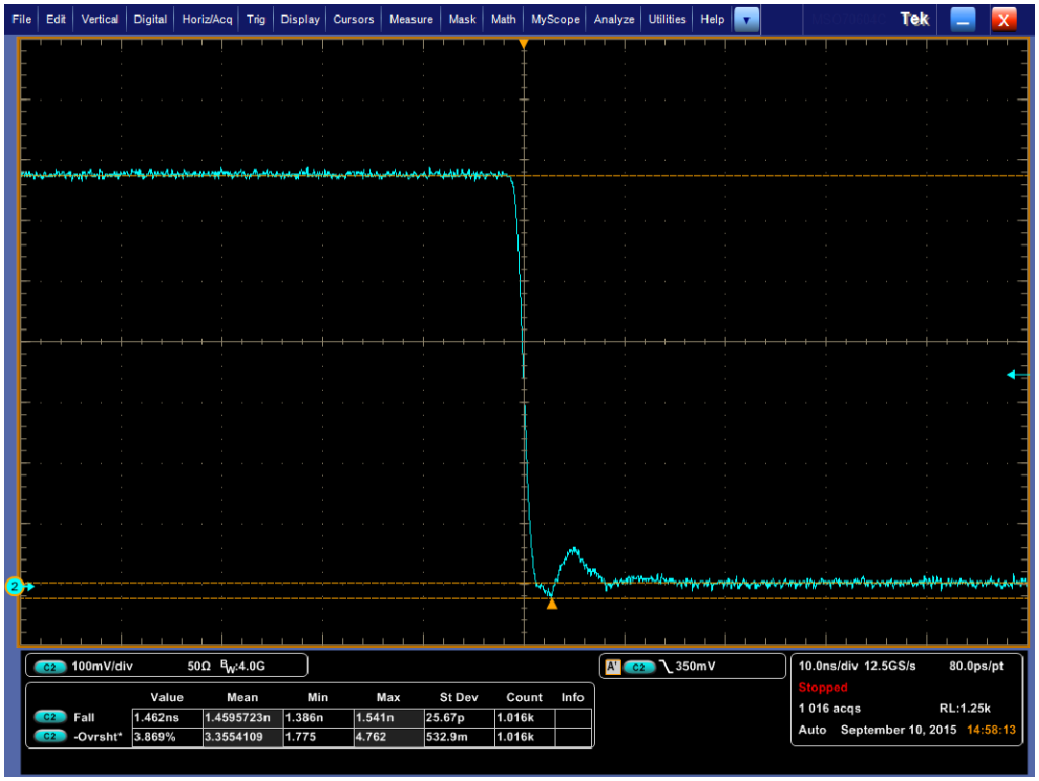
3. Analog Falling Time

Falling Time Table

10%~90%	PASS LIMITS	Mean	Minimum	Maximum	Result
Red	<=1.68ns	1.45ns	1.38ns	1.54ns	Pass
Green	<=1.68ns	1.40ns	1.33ns	1.47ns	Pass
Blue	<=1.68ns	1.47ns	1.40ns	1.58ns	Pass

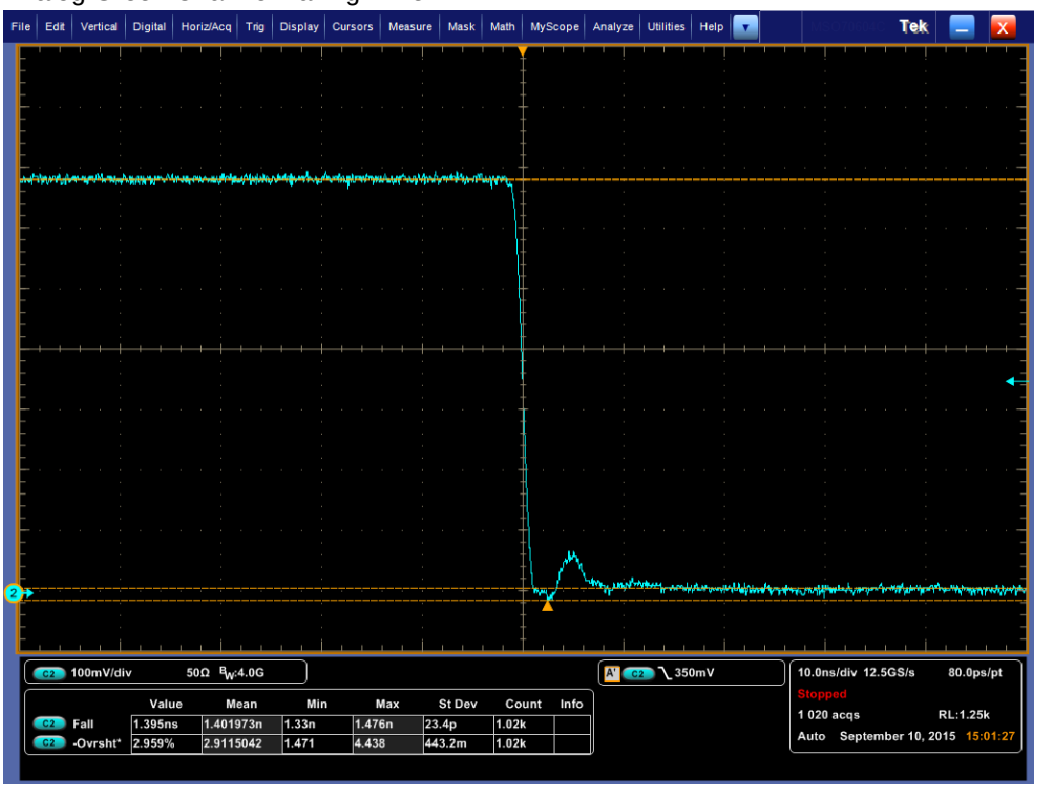
Pass Limits => 1.68ns = (1/148.500M) *25%

Analog Red Channel Falling Time

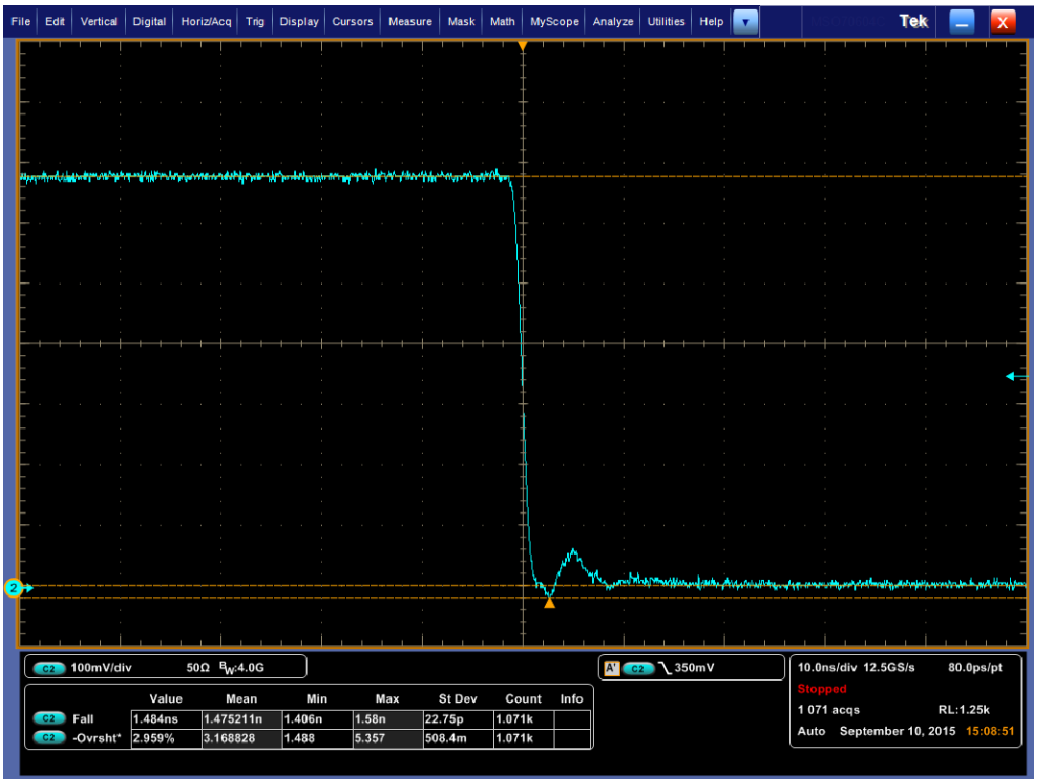


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Analog Green Channel Falling Time



Analog Blue Channel Falling Time



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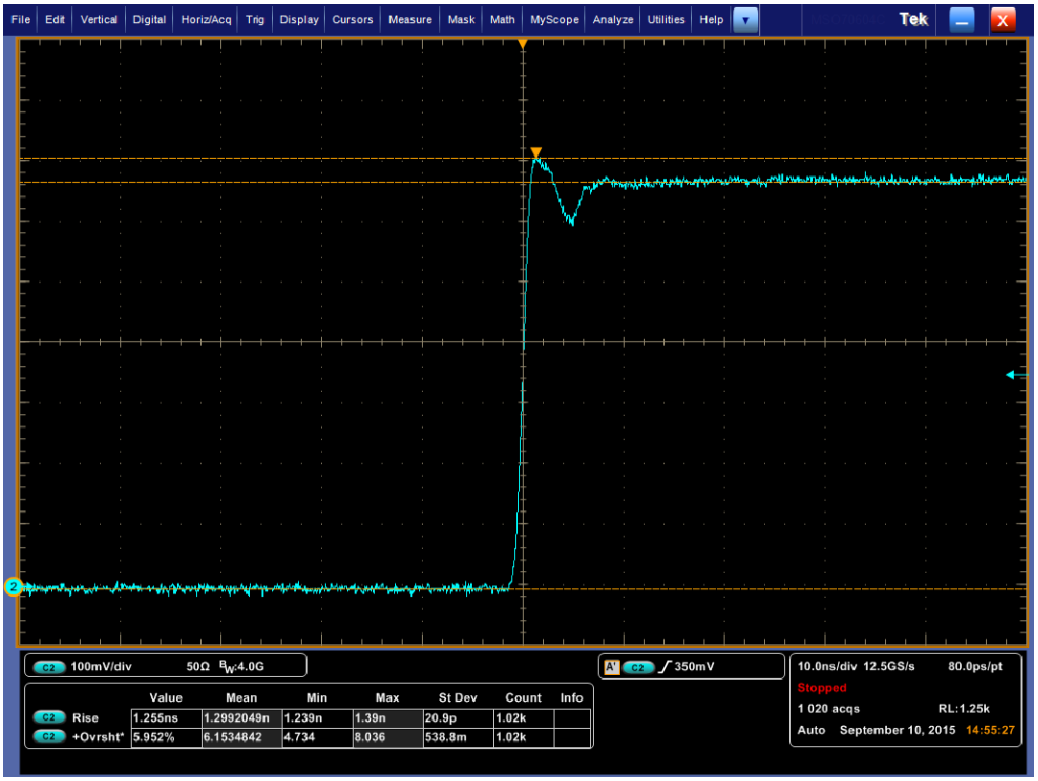
4. Analog Overshoot

Overshoot Table

Avg~Max	PASS LIMITS	Mean	Minimum	Maximum	Result
Red	<=12%	6.153%	4.734%	8.036%	Pass
Green	<=12%	8.476%	7.059%	10.06%	Pass
Blue	<=12%	6.056%	4.734%	8.036%	Pass

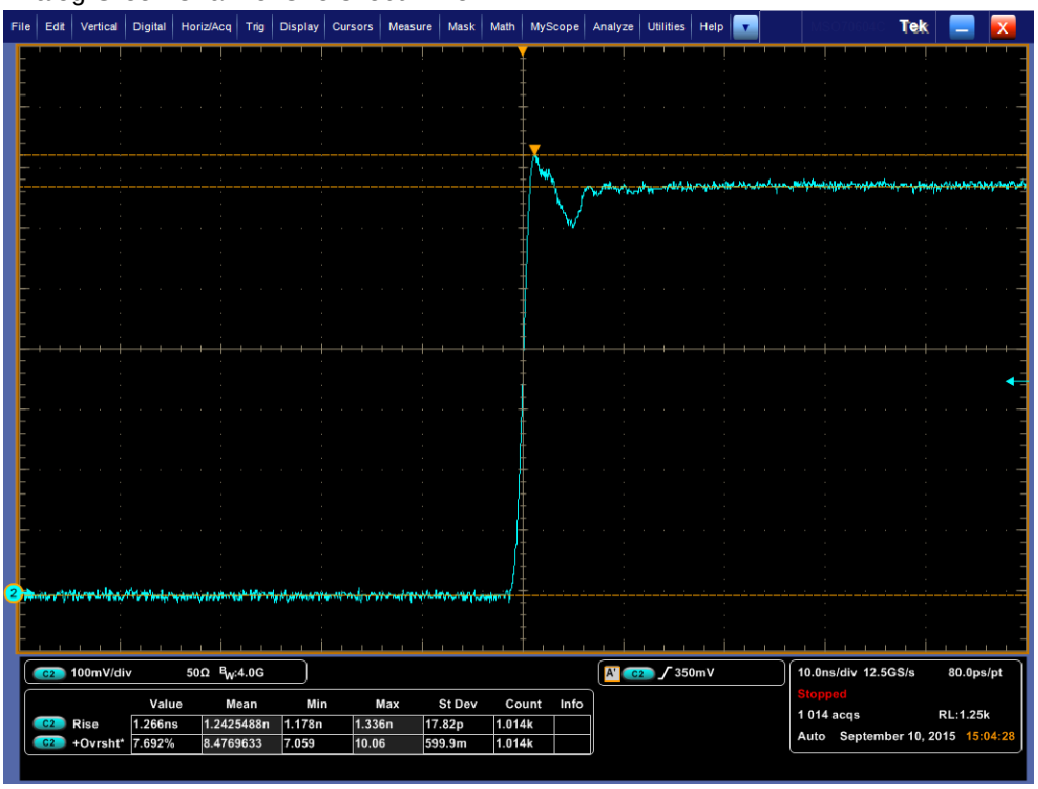
Pass Limits => 12%

Analog Red Channel Overshoot Time

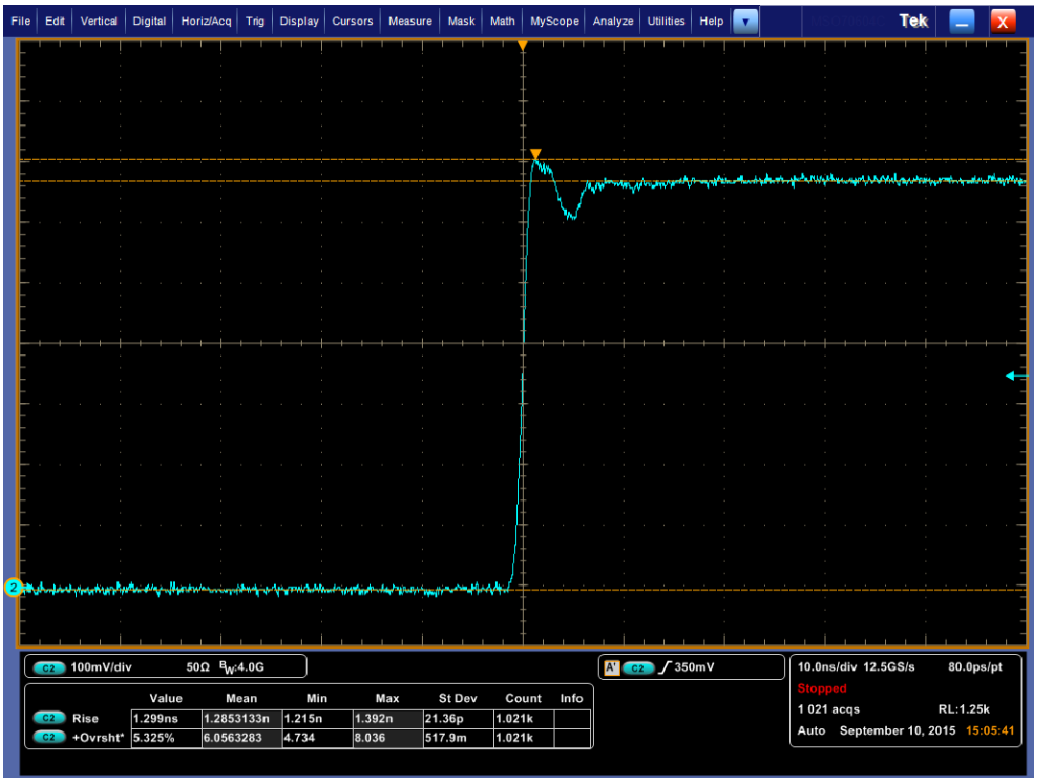


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Analog Green Channel Overshoot Time



Analog Blue Channel Overshoot Time



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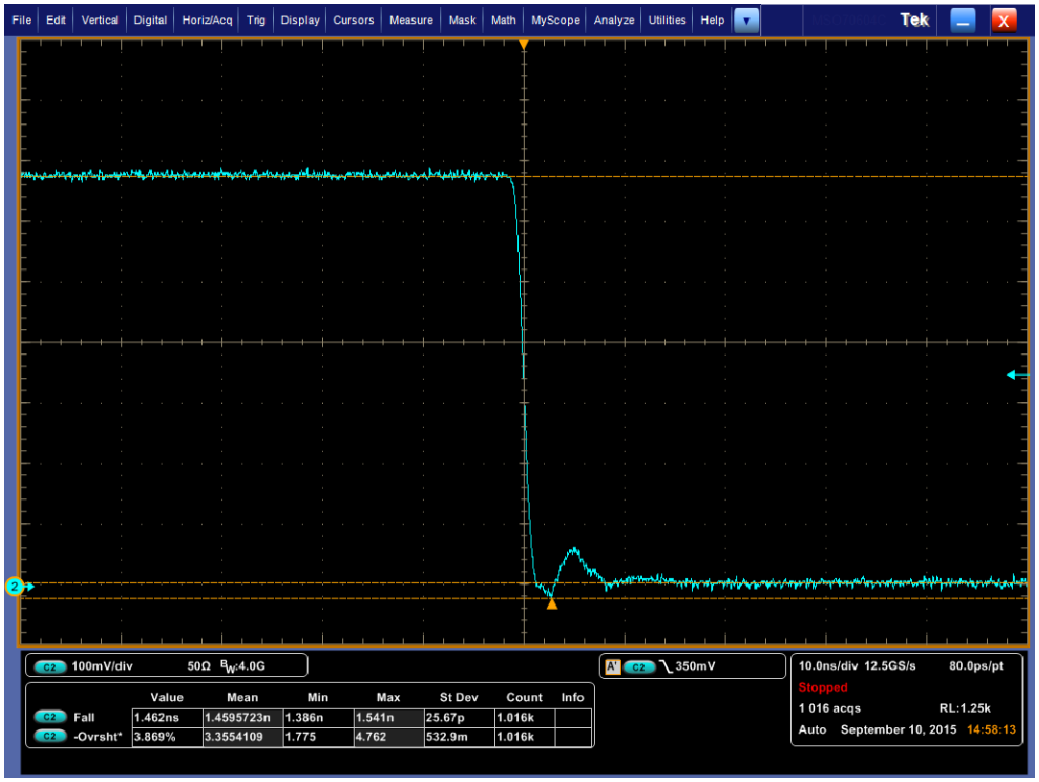
5. Analog Undershoot

Undershoot Table

Avg~Max	PASS LIMITS	Mean	Minimum	Maximum	Result
Red	<=12%	3.355%	1.775%	4.762%	Pass
Green	<=12%	2.911%	1.471%	4.438%	Pass
Blue	<=12%	3.168%	1.488%	5.357%	Pass

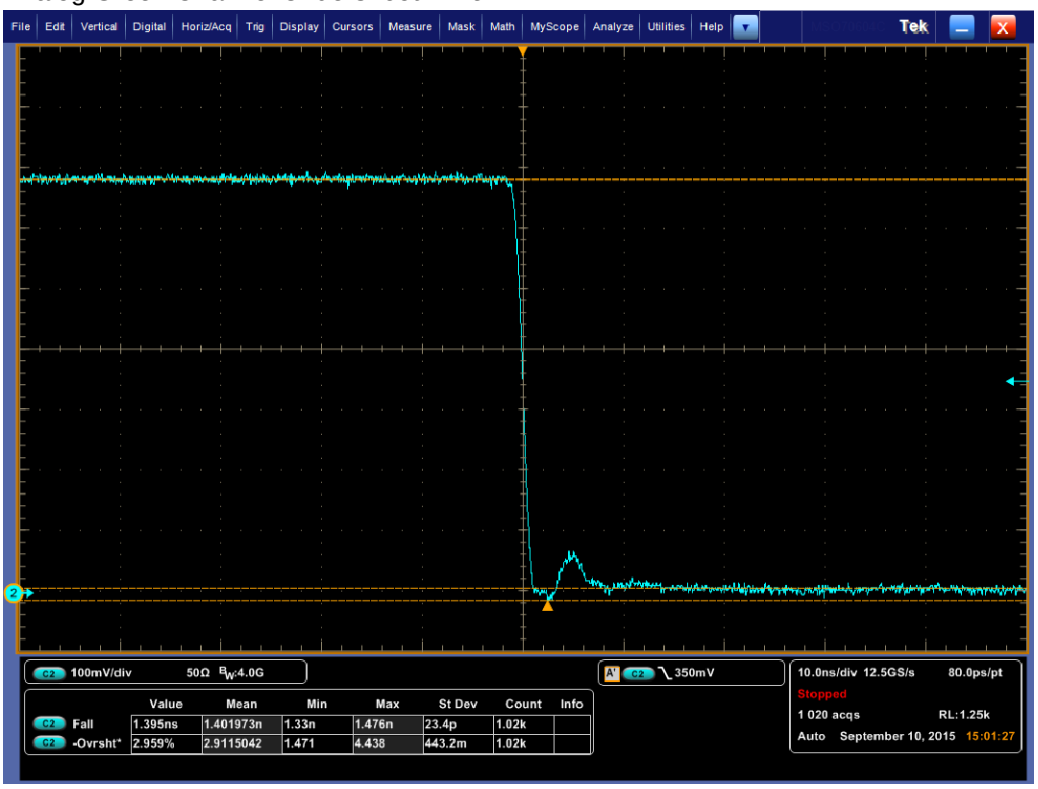
Pass Limits => 12%

Analog Red Channel Undershoot Time

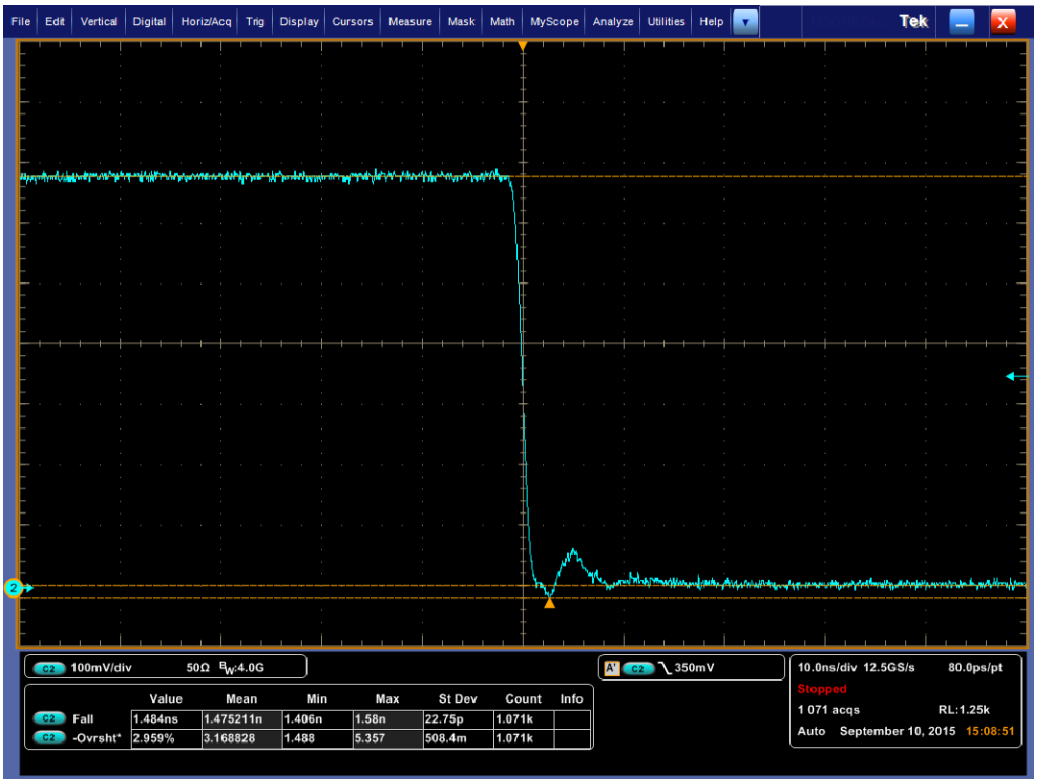


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Analog Green Channel Undershoot Time



Analog Green Channel Undershoot Time



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6. Analog Max and Min luminance voltage

Max luminance voltage

	Pass Limits	Mean	Minimum	Maximum	Result
Red	665~770mv	696.28mv	686mv	710mv	**Pass**
Green	665~770mv	704.09mv	686mv	734mv	**Pass**
Blue	665~770mv	692.47mv	682mv	710mv	**Pass**

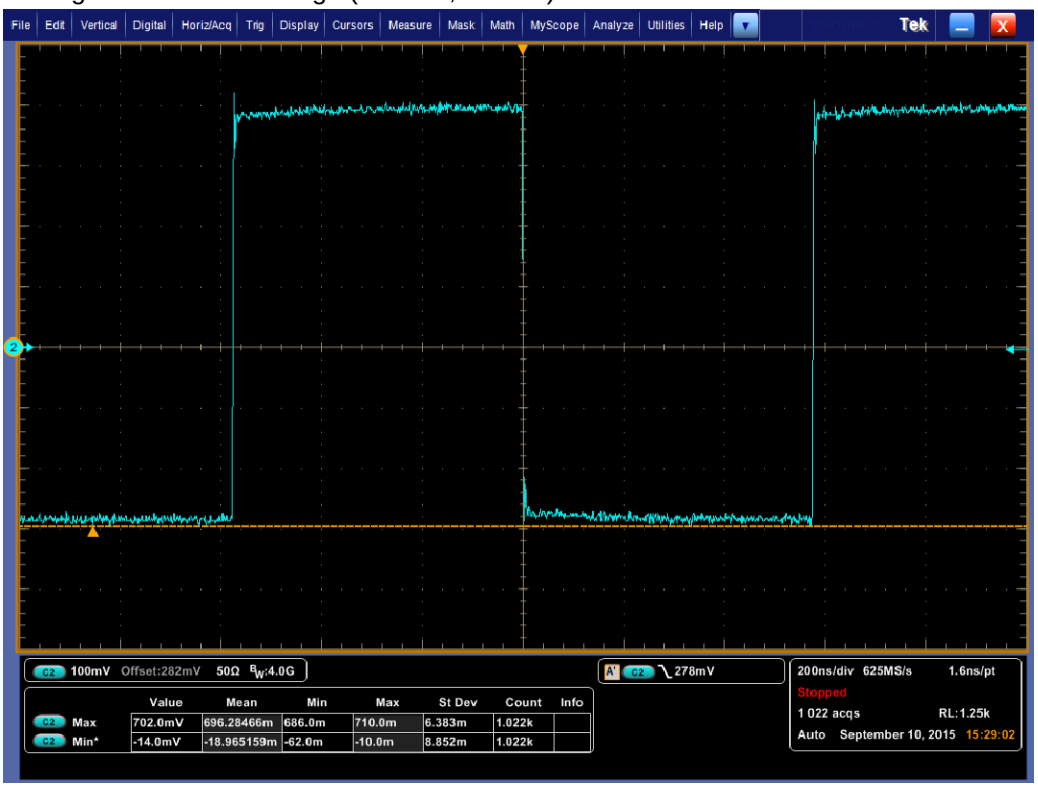
Min luminance voltage

	Pass Limits	Mean
Red	Information Only	-18.96mv
Green	Information Only	-17.54mv
Blue	Information Only	-19.37mv

Pass Limits => Max 665~770mv

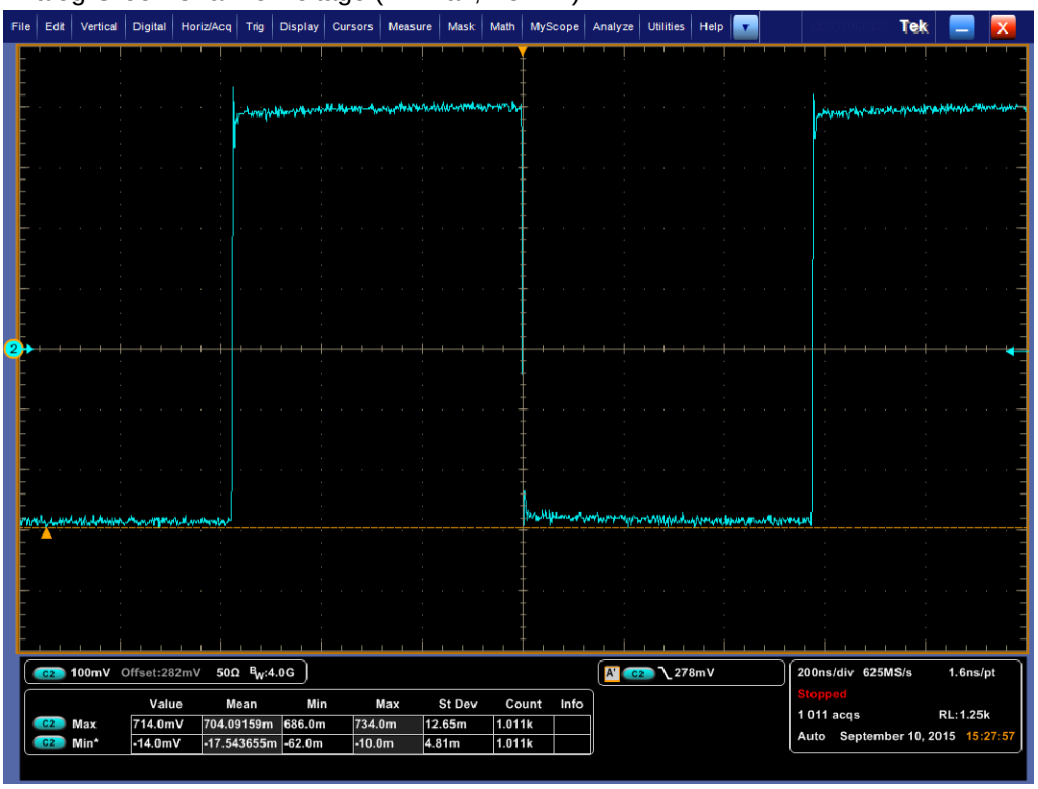
Min 0mv

Analog Red Channel voltage (P4:max, P5:min)

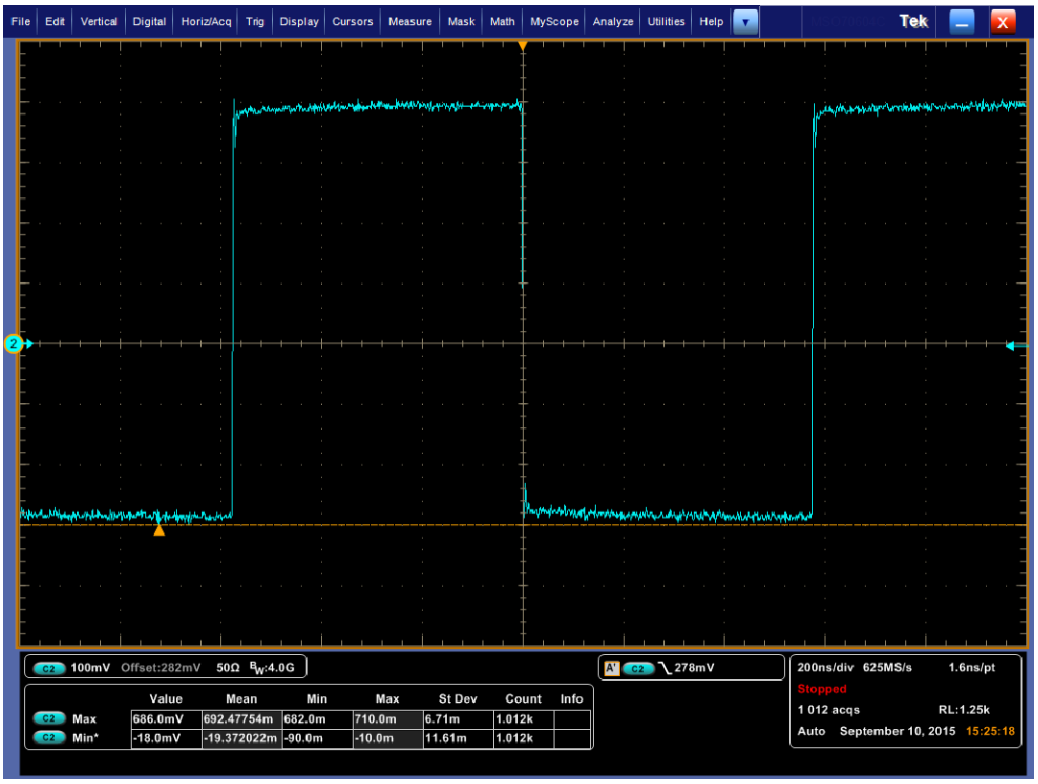


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Analog Green Channel voltage (P4:max, P5:min)



Analog Blue Channel voltage (P4:max, P5:min)



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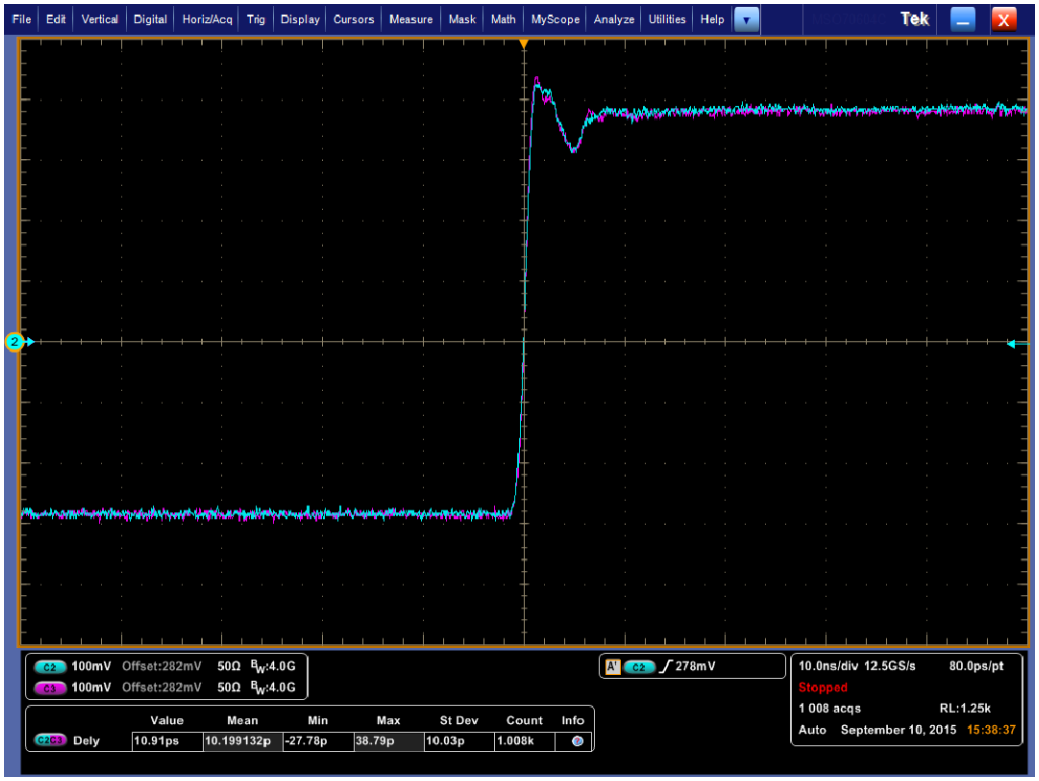
7. Analog Skewe

Skewe Channel to Channel Table

	PASS LIMITS	Mean	Minimum	Maximum	Result
Red-Green	<=3.37ns	10.199ps	-27.78ps	38.79ps	Pass
Green-Blue	<=3.37ns	-56.355ps	-90.26ps	-21.71ps	Pass
Red-Blue	<=3.37ns	-40.816ps	-70.95ps	-12.0ps	Pass

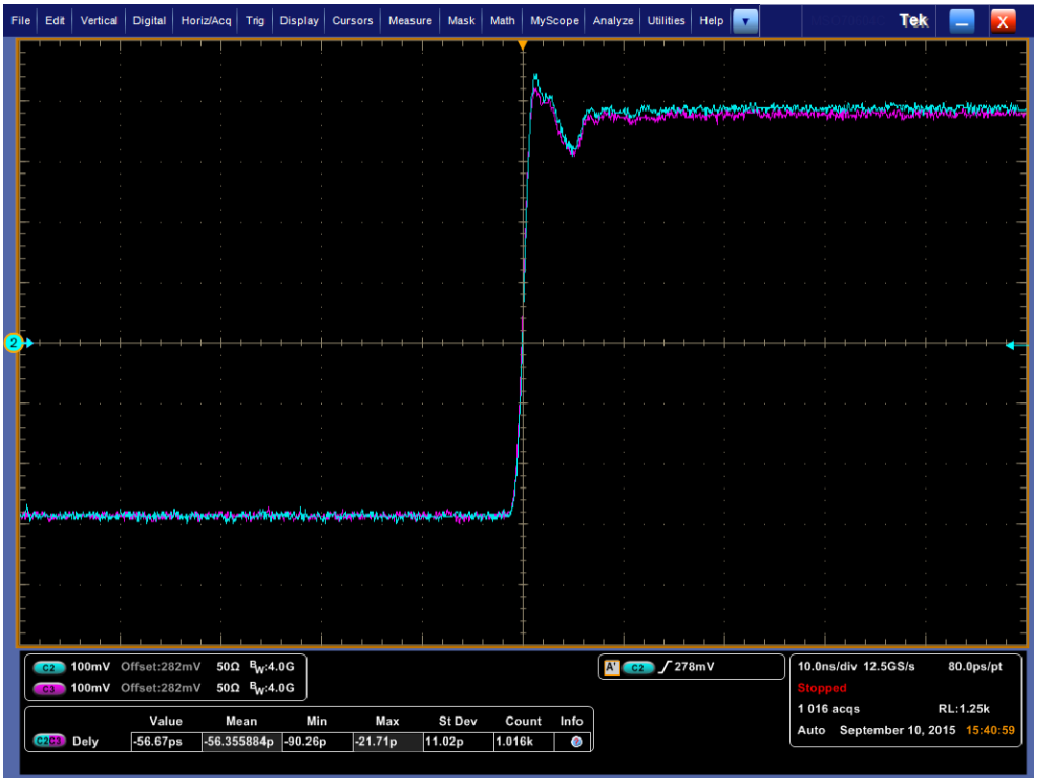
Pass Limits => 3.37ns = (1/148.500M) *50%

Analog Red-Green Channel Skewe

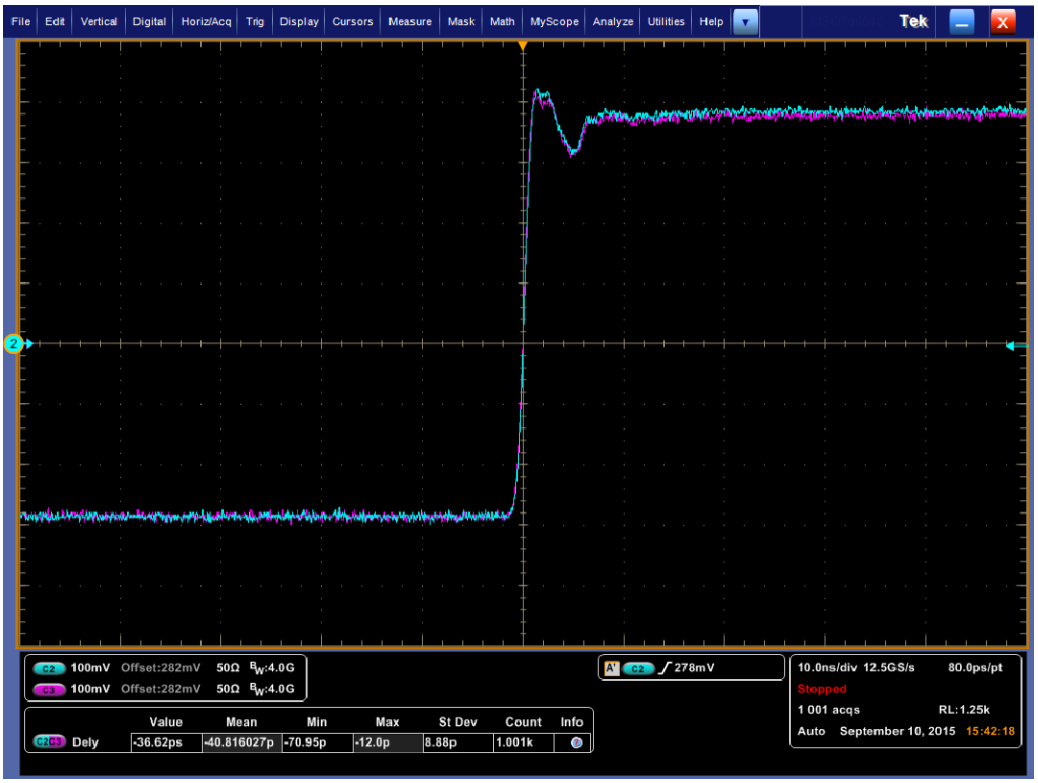


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Analog Green-Blue Channel Skewe



Analog Red-Blue Channel Skewe



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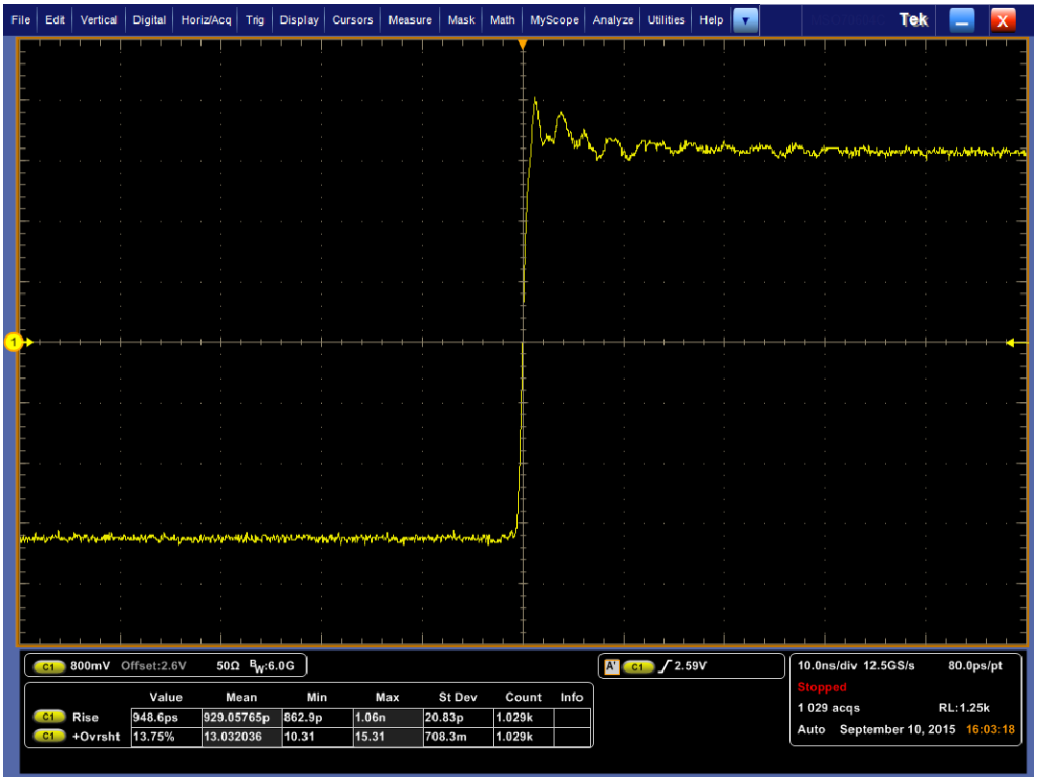
8. Digital Synchronization Rising Time

Rising Time Table

10%~90%	PASS LIMITS	Mean	Minimum	Maximum	Result
H Sync	<=5.387ns	929.05ps	862ps	1060ps	**Pass**
V Sync	<=5.387ns	1323ps	1234ps	1456ps	**Pass**

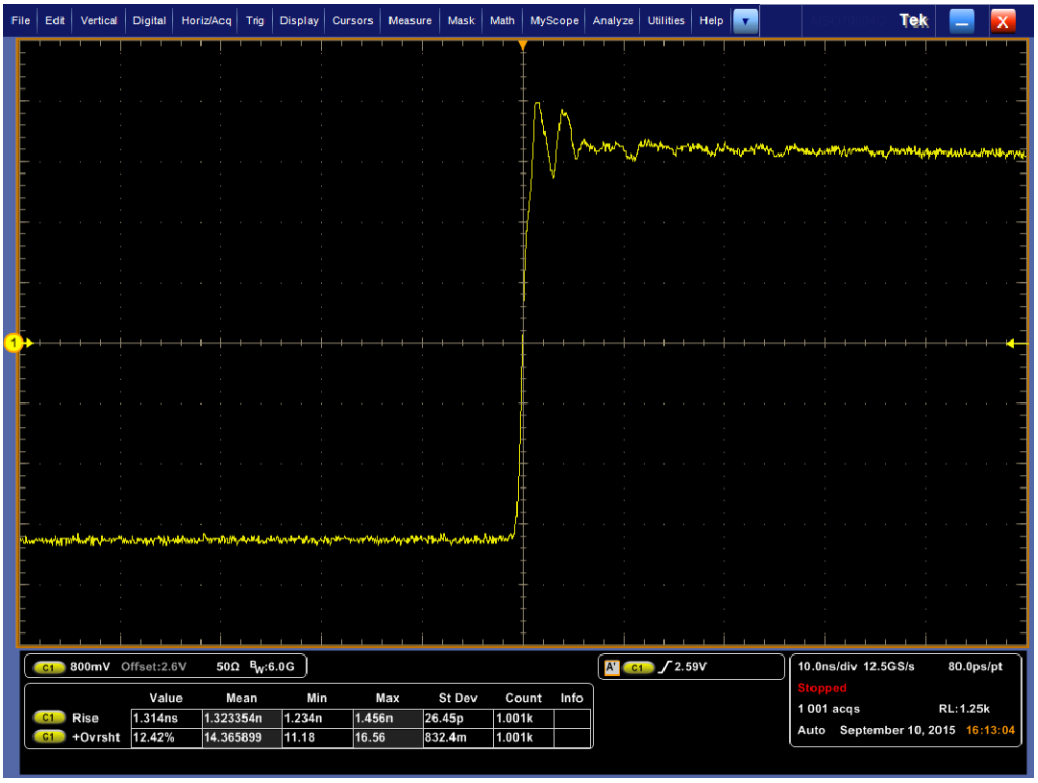
Pass Limits => 5.387ns = (1/148.500M)*80%

Digital Horizontal Synchronization Rising Time



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Digital Vertical Synchronization Rising Time



9. Digital Synchronization Falling Time

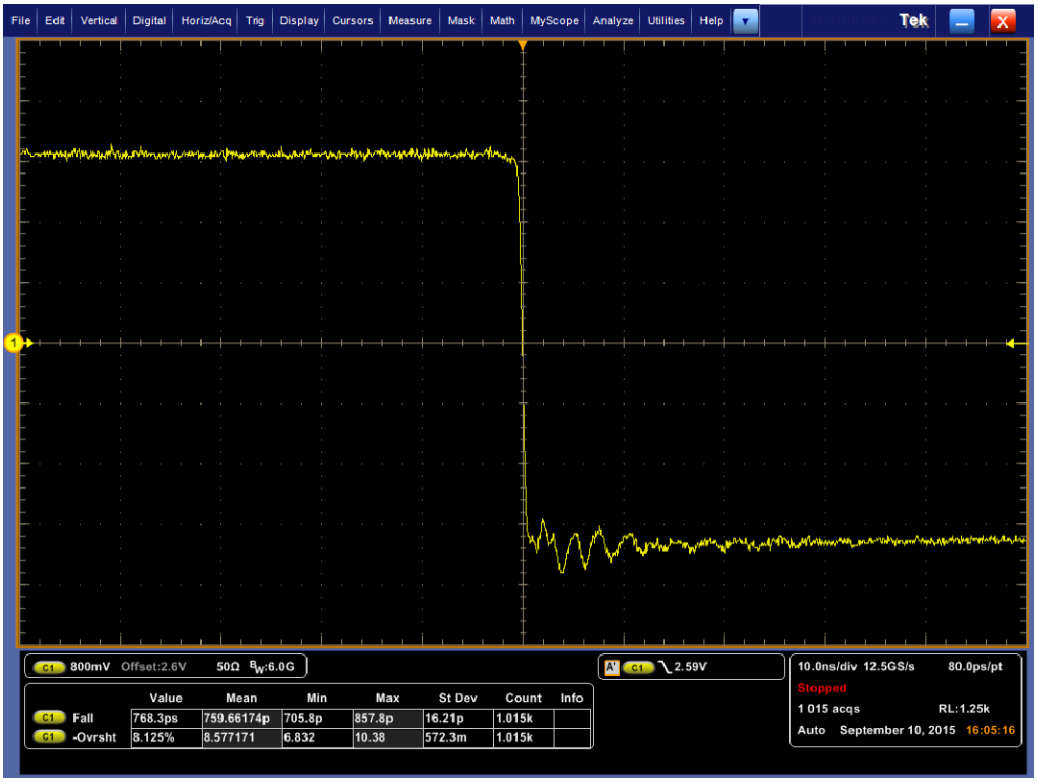
Falling Time Table

10%~90%	PASS LIMITS	Mean	Minimum	Maximum	Result
H Sync	<=5.387ns	759.66ps	705.8ps	857.8ps	**Pass**
V Sync	<=5.387ns	985.79ps	930.3ps	1053ps	**Pass**

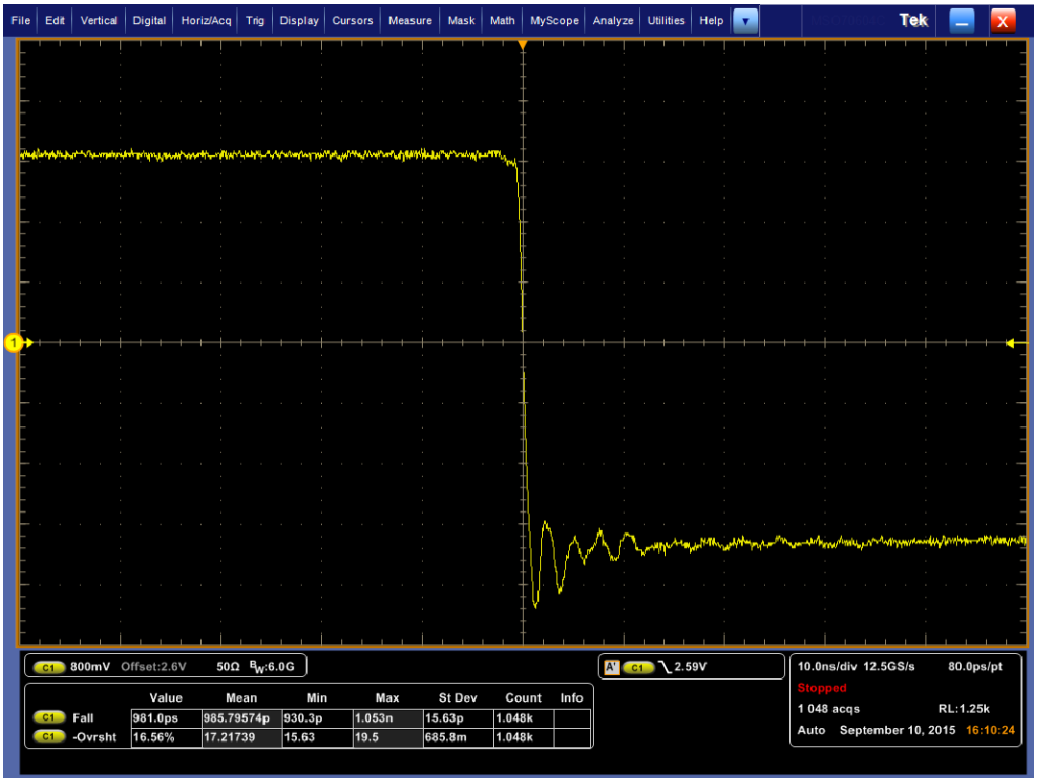
Pass Limits => 5.387ns = (1/148.500M)*80%

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Digital Horizontal Synchronization Falling Time



Digital Vertical Synchronization Falling Time



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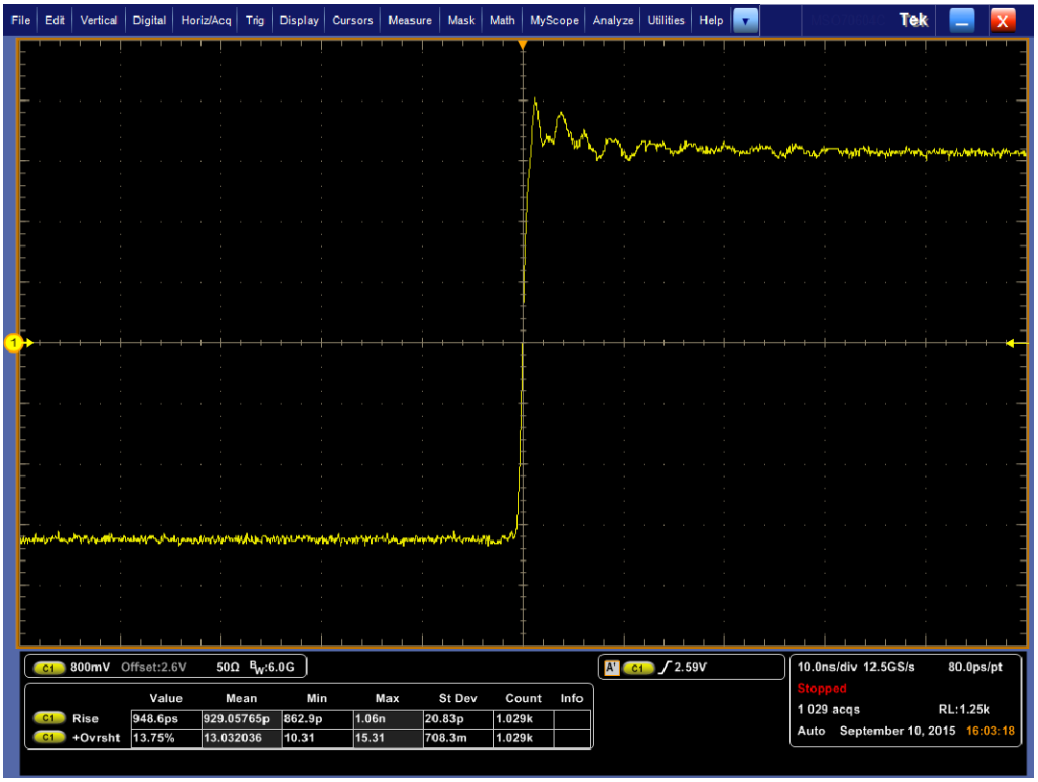
10. Digital Synchronization Overshoot

Overshoot Table

Avg~Max	PASS LIMITS	Mean	Minimum	Maximum	Result
H Sync	<=30%	13.032%	10.31%	15.31%	Pass
V Sync	<=30%	14.365%	11.18%	16.56%	Pass

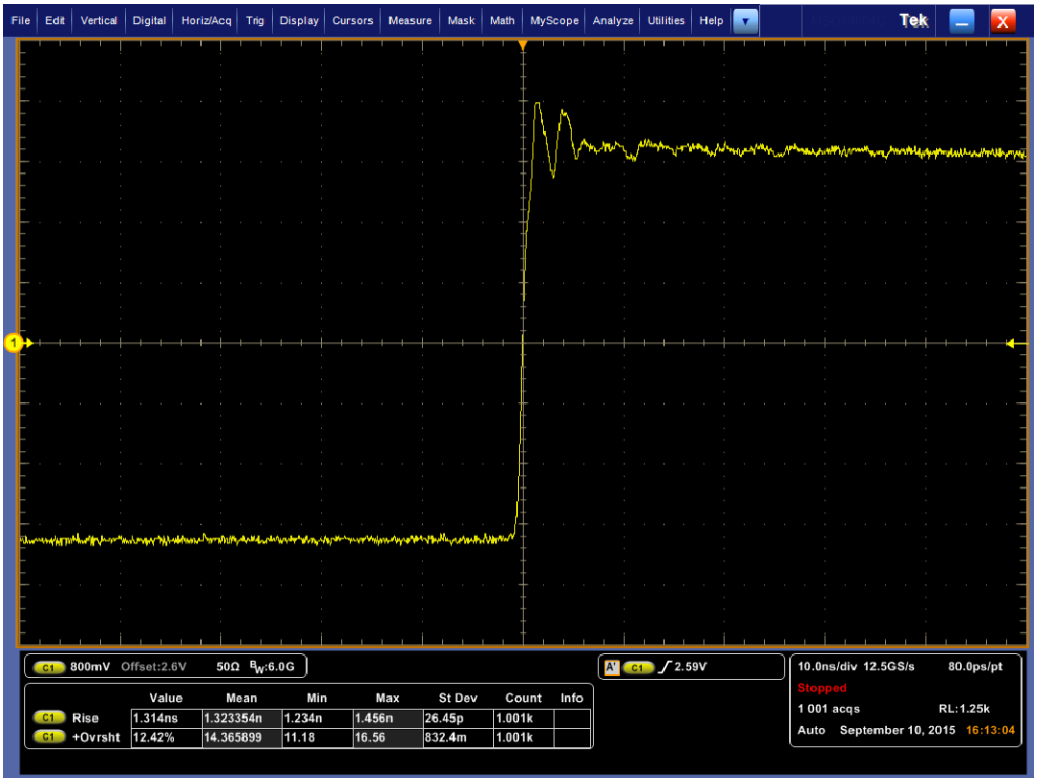
Pass Limits =>30%

Digital Horizontal Synchronization Overshoot Time



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Digital Vertical Synchronization Overshoot Time



11. Digital Synchronization Undershoot

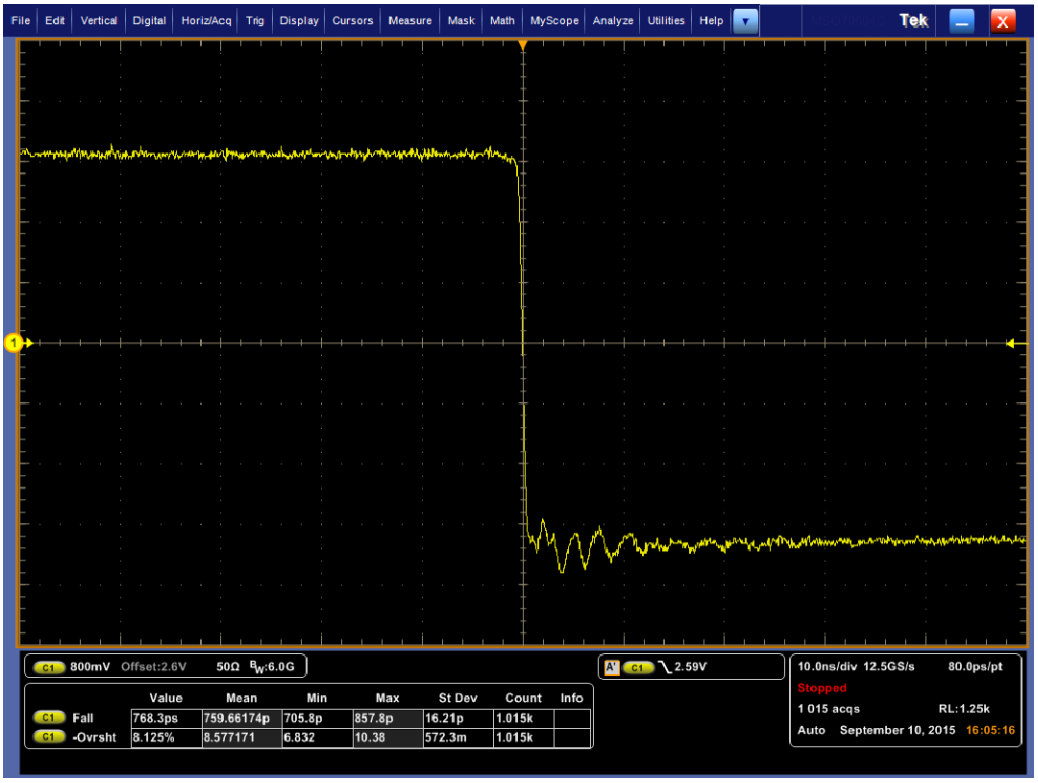
Undershoot Table

Avg~Max	PASS LIMITS	Mean	Minimum	Maximum	Result
H Sync	<=30%	8.577%	6.83%	10.38%	**Pass**
V Sync	<=30%	17.217%	15.63%	19.5%	**Pass**

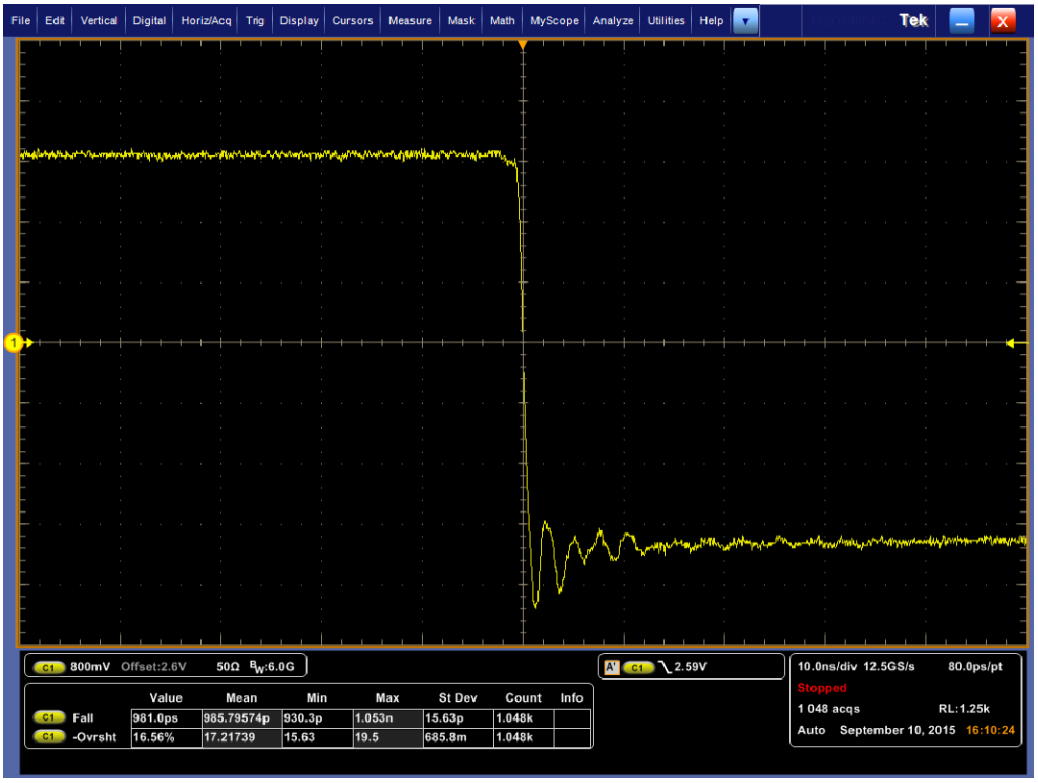
Pass Limits =>30%

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Digital Horizontal Synchronization Undershoot Time



Digital Vertical Synchronization Undershoot Time



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12. Reference

VSIS (Video Signal Standard)

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